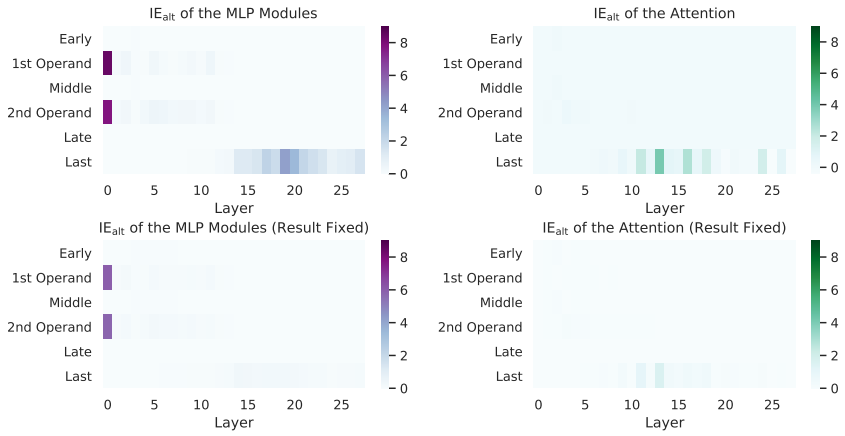

Figure 13: Indirect effect measured using the difference in the log probability as described in Equation 5 (IE_{alt}). The results are obtained with GPT-J on two-operand arithmetic queries.

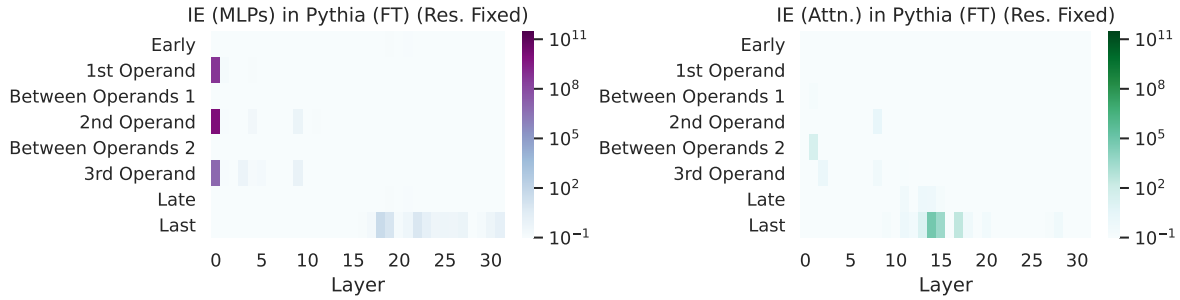

Figure 14: Indirect effect (IE) measured for the attention modules in the fine-tuned version of Pythia 2.8B on three-operand arithmetic queries, when $r = r'$.

J Additional Information Flow Visualizations

We include the IE measurements for the attention modules of GPT-J on the number retrieval task (Figure 11) and on the factual knowledge queries (Figure 12), and for Pythia 2.8B on three-operand arithmetic queries before and after fine-tuning (Figure 15). Additionally, we report the heatmap visualizations of the indirect effect measured for the following models: Pythia 2.8B (Figure 16), LLaMA 7B (Figure 17), Goat (Figure 18), and GPT-J using word numerals (Figure 19). Finally, we visualize in Figure 14 the IE of MLPs and attention modules for the fine-tuned Pythia 2.8B in the fixed-result case.

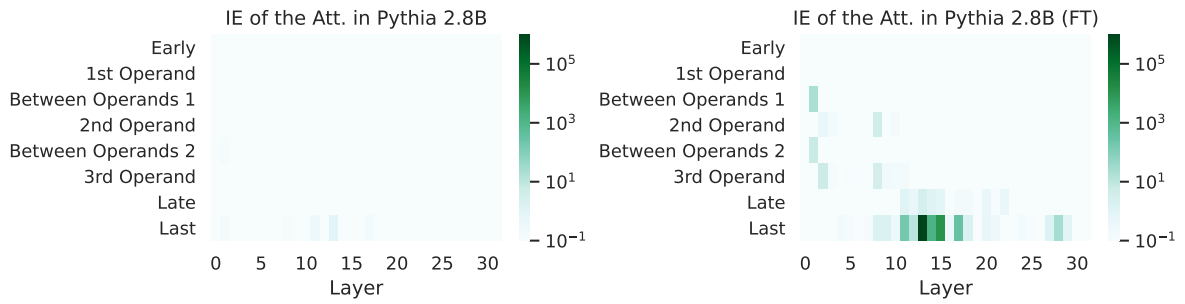

Figure 15: Indirect effect (IE) measured for the attention modules in Pythia 2.8B on three-operand arithmetic queries, before and after fine-tuning.

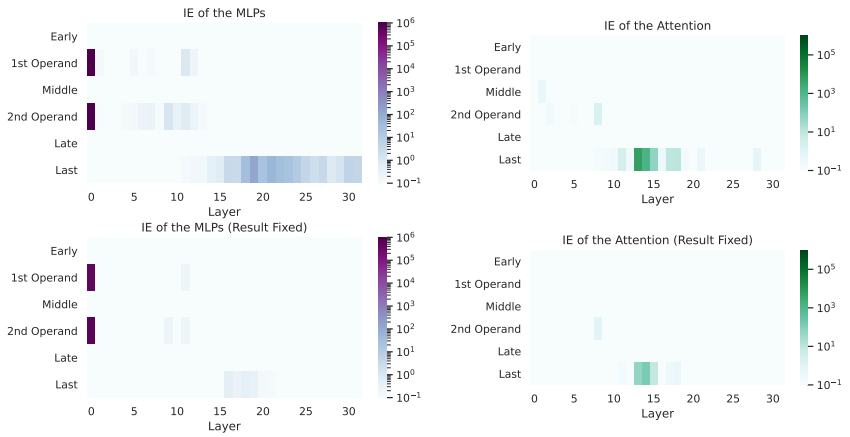

Figure 16: Indirect effect (IE) measured in the MLP and attention modules of Pythia 2.8B on two-operand arithmetic queries.

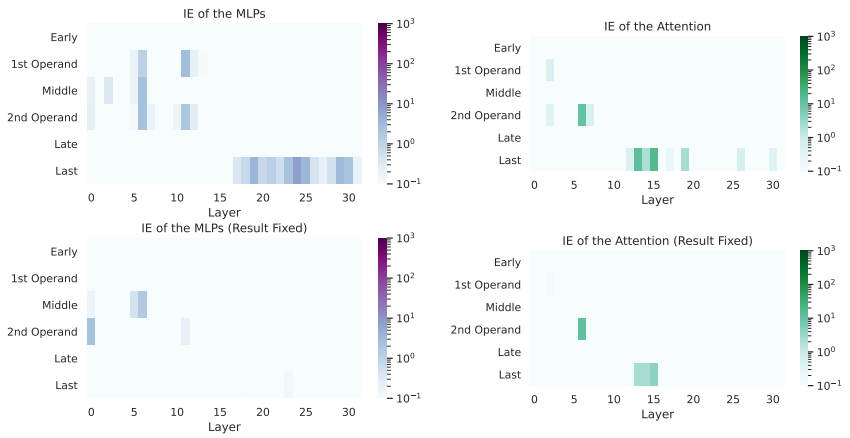

Figure 17: Indirect effect (IE) measured in the MLP and attention modules of LLaMA 7B on two-operand arithmetic queries.

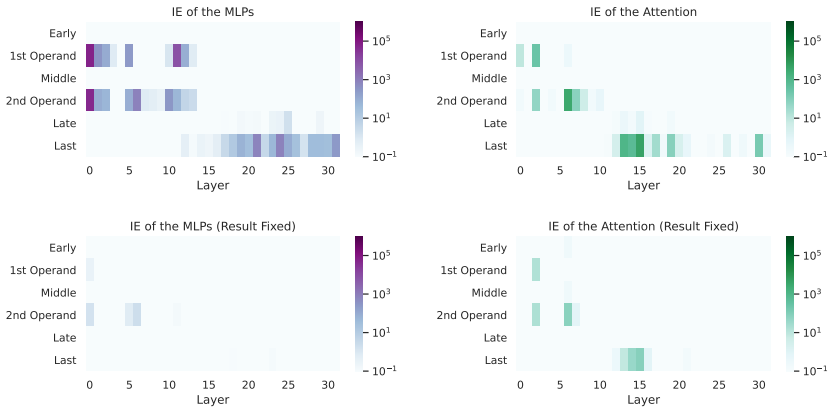

Figure 18: Indirect effect (IE) measured in the MLP and attention modules of Goat on two-operand arithmetic queries.

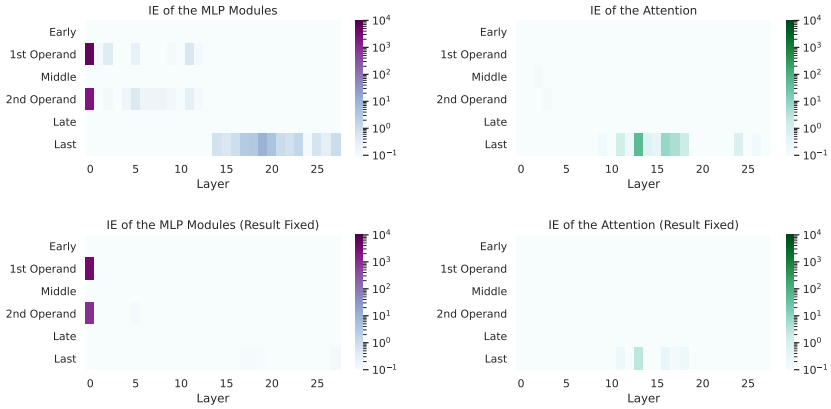

Figure 19: Indirect effect (IE) measured in the MLP and attention modules of GPT-J on two-operand arithmetic queries, using numeral words to represent quantities.